Air Quality Index Prediction and Attention Faithfulness Across Eight Indian Cities

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***Abstract*—Air quality forecasts let people and authorities act before pollution peaks, and attention-based models are often presented as showing which inputs drive those forecasts, yet this is rarely tested. This report studies daily station-level Air Quality Index (AQI) at 107 CPCB stations in eight Indian cities from 2023 to 2026. Station records are combined with satellite, reanalysis and archived forecast-weather data, ridge, GRU and XGBoost models are evaluated in locked year-level tests, and attention is audited against SHAP using input-replacement sensitivity. Every model is compared with a baseline that uses the previous day's AQI. The 28-day GRU lowered RMSE from 35.388 to 32.516 in 2025, and a weather-assisted XGBoost lowered it from 34.668 to 30.270 in 2026, with the weather forecast supplying much of that gain. Across 36 comparisons attention helped in 11, hurt in 11 and was inconclusive in 14, and attention agreed less than SHAP with measured sensitivity in all four audits. Attention maps therefore need checking before they are read as explanations. A final out-of-time evaluation opens in January 2027.**

***Index Terms*—air quality index, attention faithfulness, explainable AI, time-series forecasting, GRU, XGBoost, SHAP, cross-city transfer**

I. Introduction

A. Background and importance

Air pollution is one of the largest environmental risks to human health. The World Health Organization estimated that ambient air pollution caused 4.2 million premature deaths worldwide in 2019, that 99% of the world's population lived where its air quality guidelines were not met, and that 89% of those deaths occurred in low- and middle-income countries [1]. The State of Global Air report attributed 8.1 million deaths worldwide to air pollution in 2021 [2]. India carries a large share of this burden: the Global Burden of Disease study for Indian states attributed 1.67 million deaths in 2019, 17.8% of all deaths in the country, to air pollution, with an economic loss of US$36.8 billion, or 1.36% of gross domestic product [3].

B. Problem context and practical significance

India reports air quality through the Air Quality Index (AQI) computed from Central Pollution Control Board (CPCB) monitoring stations [4]. In this project's data, 20.8% of station-days from 2023 to 2025 fell in the Poor, Very poor or Severe bands, so roughly one station-day in five was unhealthy. A forecast issued a day ahead lets residents and authorities act before a pollution peak, but it is useful only if it beats the simplest alternative, which is to assume that tomorrow equals today, and only if users can trust what the model is said to rely on.

Two pain points follow. First, tomorrow's AQI is strongly related to today's, so a model can look accurate while adding little; in this study the best AQI-history forecaster's output correlated 0.989 with yesterday's value. Second, attention models are routinely explained by showing their attention weights, but it has not been established that those weights reflect what drives the output of an AQI regression model.

C. Research gap

The critical gap is that attention is adopted as an explanation without a fixed-model test. The reviewed air-quality studies report predictive gains and component ablations for attention [5], [6], [7] and compare explanation methods on classification data or a single monitoring network [8], [9], [10], but within the bounds of our survey [11] none combines a fixed-model attention faithfulness test, grouped realistic perturbations and consistency across cities for station-level multimodal AQI regression. The gap matters because accuracy cannot reveal a misleading explanation: a user may credit wind or satellite signals to a model that ignores them. The debate over attention as explanation arose in text classification [12], [13], and its answer for continuous, correlated environmental inputs is unknown. A second roadblock is evaluation practice, since the reviewed studies differ in splits, targets and horizons, and in one the

independence of the split from preprocessing is unverified [14].

D. Research questions

1) How much do a history-based model and a weather-assisted model improve next-day station AQI forecasts over the previous-day baseline in locked chronological tests?
2) Does attention improve accuracy over uniform-weight controls, and does the answer hold across input sets and test designs?
3) Do attention rankings agree with measured input-replacement sensitivity as well as SHAP rankings do?
4) Which inputs account for the forecast skill beyond the baseline?

E. Research objectives

1) To build, by September 2026, an aligned daily dataset for 107 CPCB stations in eight cities with a verified end-of-day AQI target, satellite and weather inputs and documented quality flags.
2) To evaluate ridge, GRU and XGBoost forecasters in three locked tests (2024, 2025 and January–September 2026), each scored once against the baseline with 95% paired block-bootstrap intervals.
3) To compare 36 attention variants with uniform controls on two input sets under chronological, leave-one-city-out and season-block designs.
4) To audit attention and SHAP rankings against replacement sensitivity on 300 sampled rows per checkpoint.
5) To quantify, with input ablations and importance scores, how much forecast skill comes from yesterday's AQI, older history, calendar and location, and weather forecasts.
6) To publish the saved results in a dashboard that separates same-day estimates, backtests and the pending out-of-time evaluation.

F. Research contribution

This work makes four contributions. (1) It provides three locked, independently verified forecast evaluations against an explicit previous-day baseline across eight Indian cities. (2) It gives paired evidence on 36 attention variants showing that the accuracy effect of attention depends on the input set and test design. (3) It audits attention against SHAP with replacement sensitivity and finds lower agreement for attention in every sampled audit. (4) It shows how little of the forecast skill comes from AQI history beyond yesterday's value and how much comes from weather forecasts, with a reproducible code and dashboard package.

G. Paper roadmap

Section II reviews related work and states the research gap. Section III presents the methodology. Section IV reports results and discussion, Section V concludes, Section VI describes the study limitations and Section VII suggests future scope.

II. Related Work

A. Classification of approaches

The ten reviewed studies [11] fall into three overlapping groups, and the attention debate and post-hoc explanation methods form their background. Attention was introduced as a soft alignment mechanism [15], and it was later shown that attention weights can be altered without changing predictions [12], although the conclusion depends on how an explanation is defined and tested [13]. Post-hoc attribution such as SHAP assigns each input a contribution to a prediction [16], but explanation methods can disagree on the same model [17].

Explanation evaluation and attribution studies tested explanations against prediction changes [8], [9] or made grouped attribution efficient [18]. *Attention-based prediction and spatial assessment* studies used attention for pollutant or AQI prediction [19], [10], [14], [6], [7]. *Transfer learning and spatial generalization* studies examined prediction at sites with little data [5], [20]. Scores in these papers use different targets, horizons, data and splits, so they cannot be ranked against one another, and AQI differs from PM2.5 and PM10 concentrations.

TABLE I
COMPARATIVE SUMMARY OF THE REVIEWED STUDIES

Study	Year	Method	Key contribution	Study limitation
Turbé et al. [8]	2023	Compared six attribution methods on CNN, BiLSTM and Transformer classifiers with corrupted blocks	Corruption-based audit that ranks attribution methods	Transfer to regression remains unvalidated.
Šić et al. [9]	2025	Tested 23 perturbation methods and 12 attribution methods on five UCR datasets	Showed that attribution rankings depend on the perturbation design	The metrics are limited to classification probabilities.
Ma et al. [5]	2024	Pretrained an adaptive bidirectional GRU with temporal and external attention on nine Beijing stations, then fine-tuned on the target site	Two-stage attention for sites with little data	Adaptation requires target labels.
Li et al. [19]	2023	Trained a physics-inspired graph network with attention on 1,604 stations in China	Combined transport physics with learned spatial relations (site-based PM10 R^2 about 0.85)	The physical meaning of the learned coefficients is limited.
Zhang et al. [10]	2026	Trained an explainable Transformer (X2-AQFormer) for NO_x and PM10 in Stockholm up to 72 hours ahead	Intrinsic explanations compared with GradientSHAP	The evaluation is confined to the Stockholm network.
Liu et al. [14]	2025	Combined a Transformer and a BiLSTM to predict daily AQI in three northern Chinese cities	Daily AQI forecasting with SHAP feature analysis	The preprocessing description does not establish that the split is independent of fitting.
Poelzl et al. [20]	2025	Compared predictors, generalization and adaptation (TrAdaBoostR2) for PM10 between Graz and Zagreb	Separated generalization from label-assisted adaptation	The experiment covers two cities and excludes exceptional events.

Hertel et al. [18]	2026	Introduced SHAPformer, a masked-attention Transformer, on electricity data	Sampling-free SHAP for grouped inputs	The cost grows exponentially with the number of feature groups.
Chen et al. [6]	2023	Proposed STA-ResConvLSTM for PM2.5 around Beijing	Spatial–temporal attention in residual blocks	The model is limited by abrupt changes and by horizons beyond 48 hours.
Liu et al. [7]	2023	Developed STAA-GCN for PM2.5 on Beijing, Tianjin and London datasets	Adaptive graph attention over station relations	Prediction in unmonitored areas is outside the evaluation.

B. Gap synthesis

The reviewed literature shows that attention is widely used for air-quality prediction, but the evidence for it comes in three kinds that are rarely combined: predictive gains and component ablations [5], [6], [7]; comparisons of explanation methods on classification data [8], [9] or one monitoring network [10]; and transfer experiments that mix zero-shot, station-holdout and label-assisted settings [5], [19], [14], [20], [7]. This leaves four gaps. The methodological gap is that no study in the reviewed set tests attention against input-replacement sensitivity on a fixed model for station-level multimodal AQI regression. The knowledge gap is that the attention-as-explanation debate has not been applied to correlated environmental inputs. The population gap is that each study covers one or two cities, so consistency across regimes is unknown. The empirical gap is that conflicting views on attention faithfulness have no resolution for this task. This is a bounded finding of our survey and not proof that no such study exists. This study addresses these gaps by pairing locked forecast evaluations with fixed-model audits in eight cities (Table II).

TABLE II
MAPPING OF RESEARCH GAPS TO OBJECTIVES AND CONTRIBUTIONS

Gap type	Addressed by	Objective	Section
Methodological	Fixed-model replacement audit of attention versus SHAP	4	III-H, IV-E
Knowledge	36 paired attention-versus-uniform comparisons on environmental inputs	3	IV-D
Population	Eight cities, leave-one-city-out and season-block designs	1, 3	III-A, IV-D
Empirical	Baseline comparison and input ablations on locked tests	2, 5	IV-A to IV-C

III. METHODOLOGY

A. Dataset description

Hourly station AQI came from CPCB exports [4]. Satellite inputs were Sentinel-5P NO_2 and HCHO [21] and MODIS aerosol optical depth [22], and weather inputs were ERA5-Land daily reanalysis [23], all extracted through Google Earth Engine on a 0.03° grid. The 2026 weather model also used GFS forecasts issued 24 hours before each valid hour, taken from the Open-Meteo Previous Runs API [24]. CPCB is the national agency for the official index, and the satellite and reanalysis products cover every city, which made one consistent pipeline possible. The 107 stations are in Delhi (39), Hyderabad (14), Mumbai (14), Bengaluru (13), Chennai (8), Kolkata (7), Jaipur (6) and Lucknow (6), covering plains, coastal, upland, semi-arid and delta climates (Table III). A further 20,086 station-days from January to September 2026 at 91 stations form the weather test.

TABLE III
ALIGNED DATA BY YEAR

Year	Station-days	With AQI	Stations
2023	31,813	30,081	107
2024	33,860	31,722	106
2025	34,189	32,842	107
All	99,862	94,645	107

The target is skewed. Of 93,815 usable station-days, 12.4% were Good (AQI 0–50), 35.1% Satisfactory (51–100), 31.6% Moderate (101–200), 11.0% Poor (201–300), 7.4% Very poor (301–400) and 2.4% Severe (401–500). The mean was 138.4, the median 105 and the standard deviation 98.4.

B. Preprocessing

1) *Time alignment.* CPCB's hourly AQI is a trailing 24-hour average, so a daily mean is blurred across two days: its correlation with the previous day's satellite NO_2 (0.217) nearly matched that with the same day's (0.230). The target is *AQI_eod*, the hourly AQI at 05:00 IST on the following day, whose window covers almost exactly the UTC day described by the inputs. All dates are UTC.
2) *Cleaning.* Days on which AQI stayed constant below the 500 cap were flagged as stuck-sensor readings (906 rows) and excluded. Missing targets were never imputed. An independent audit reproduced all 31,813 aligned 2023 rows from the raw files with no mismatches.
3) *Missing inputs.* Satellite values are kept as missing under cloud. Tree models read missing values natively; ridge used median imputation with missing indicators; the GRU received a validity channel and zeros after normalization.
4) *Normalization.* Inputs were standardized with the training-window mean and standard deviation (z-score), which suits gradient-trained networks and keeps the scale of each input comparable. The GRU target was standardized the same way.
5) *Encoding.* Day of year was encoded cyclically as sine and cosine so that 31 December and 1 January are close. City and station identity are not inputs; location enters only as latitude and longitude.
6) *Balancing.* No resampling was used because AQI is continuous and resampling would distort the target. The effect of the skew is instead exposed by reporting errors per AQI band.

C. Feature engineering and selection

Table IV lists the input groups. For next-day forecasts every input for target day D uses only observations from days up to D−1, and weather forecasts were issued before D began.

Three feature families were compared for the tree and linear models: a short family of 9 inputs, a long family of 45 and a delayed-weather family that adds ERA5 values lagged 7 and 14 days (57 inputs). There was no importance-based pruning; the family and model were chosen by lowest RMSE on October 2024, a month held out from fitting (Table V).

TABLE IV
INPUT GROUPS

Group	Inputs	Count	Used by
Past AQI	Previous 28 daily AQI values	28	GRU, XGBoost, ridge
AQI summaries	Mean, spread and valid-day count over 3, 7, 14, 28 days; change from the day before	13	XGBoost, ridge
Calendar	Day of year (sine, cosine)	2	All models
Location	Station latitude, longitude	2	All models
Forecast weather	GFS temperature, humidity, wind speed, rain, pressure, issued 24 h ahead	5	Weather XGBoost
Satellite and ERA5	NO_2, HCHO, aerosol; temperature, dew point, wind, pressure, rain (same day)	n/a	Same-day ridge, attention study

TABLE V
MODEL SELECTION ON OCTOBER 2024 (2,642 ROWS)

Candidate	Inputs	RMSE
GRU-28days	28-day window	31.501
Ridge-short	9	33.483
XGBoost-delayed ERA5	57	33.624
RandomForest-delayed ERA5	57	34.036
RandomForest-long	45	34.350
Ridge-delayed ERA5	57	34.504
Persistence	yesterday only	34.546
Ridge-long	45	34.673
XGBoost-long	45	34.881
XGBoost-short	9	35.358

D. Algorithm selection and justification

The baseline repeats the previous day's AQI, which every candidate must beat. Ridge regression is a transparent linear reference. XGBoost [25] fits non-linear trees, handles missing values and was trained on the change from yesterday so that the baseline is its default answer. The GRU [26] is a recurrent network that reads a 28-day window and can use temporal patterns that lag features capture only partly. A three-seed GRU ensemble lowers seed variance. In Table V the GRU had the lowest RMSE on the selection month, ahead of all tree and linear candidates, and it was frozen for the 2025 test. For 2026 a second candidate added archived weather forecasts to XGBoost. The attention study used GRU encoders with temporal, feature or spatial attention because these are the mechanisms whose weights are read as explanations.

E. System workflow

Fig. 1 shows the workflow from raw data to dashboard. The chronological split is the central safeguard: the model is fitted, selected and calibrated before the test period is scored once.

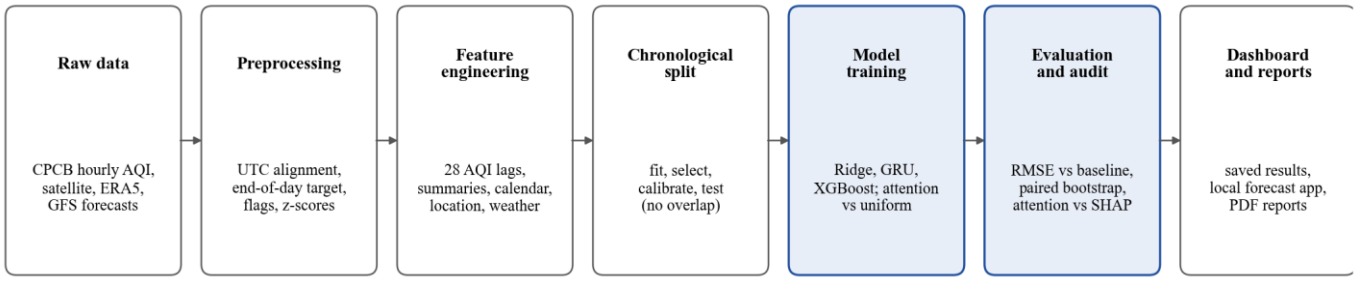


Fig. 1. System workflow from raw data to dashboard. The shaded stages are the study's two main experiments.

F. Experimental setup

TABLE VI
HARDWARE, SOFTWARE AND PARAMETERS

Item	Value
Computer	AMD Ryzen 7 7840HS (8 cores, 16 threads), 15.3 GB RAM, Windows 11 Home
GPU	NVIDIA GeForce RTX 4060 Laptop, 8 GB, driver 592.82, CUDA 12.8
Software	Python 3.13.3; PyTorch 2.10.0+cu128 [27]; XGBoost 3.4.1; scikit-learn 1.6.0 [28]; pandas 2.2.3; NumPy 2.1.2; SciPy 1.14.1; SHAP 0.53.0
GRU	1 layer, 64 units; Adam, learning rate 2×10^{-3}, weight decay 10^{-4}; batch 256; at most 60 epochs with early stopping (patience 8) on a 15% date-based validation split; gradient clip 1.0; MSE on the standardized target; 3 seeds (stopped at 23, 19 and 26 epochs)
XGBoost	600 trees, depth 5, learning rate 0.035, row sample 0.85, column sample 0.9, L2 5, minimum child weight 15, histogram method on GPU, seed 2
Ridge	alpha 1.0 after median imputation, missing indicators and standardization
Run time	Development grid for 2025 (all candidates, GPU): 4 min 21 s. CPU inference: GRU ensemble 65 ms and XGBoost 9 ms per 1,000 station-days

G. Evaluation protocol

Each forecast test fixed its model, selection and interval radius before the test period was scored once (Table VII), and the dataset hashes were checked before scoring. The three tests use different periods and stations, so a model is compared only with the baseline on its own rows. The attention study used three designs on 2023 data: chronological (train January–August, validate September–October, test November–December), leave-one-city-out in eight folds, and season-block in four folds of interleaved months. The first attention runs stopped early on randomly held-out weeks; every attention result here comes from corrected runs with purged date-based validation.

TABLE VII
DESIGN OF THE THREE LOCKED FORECAST TESTS

Step	2024 same-day	2025 next-day	2026 weather
Fit	Jan–Sep 2023	Jan 2023–Sep 2024	Jan 2024–Sep 2025
Select	Oct 2023	Oct 2024	Oct 2025
Calibrate	Nov–Dec 2023	Nov–Dec 2024	Nov–Dec 2025
Scored	2024	2025	Jan–Sep 2026
Rows	28,803	30,263	20,086

The metrics are root-mean-square error (RMSE), mean absolute error (MAE) and R^2. Gains are RMSE differences between paired models on identical rows, with 95% intervals from paired calendar-week block bootstraps (2,000 draws for the 2026 test). The intervals cover test-week sampling only and are unadjusted across multiple comparisons. A 90% prediction range was set as a fixed radius from calibration errors, and its coverage was checked on the test. A candidate could replace the default model only if the lower bound of its paired gain was positive against both the baseline and the GRU, with no loss of sharp-change precision or recall at a 50-point threshold. The rule was set before the 2026 test.

H. Faithfulness audit

Two audits were run on frozen models. In the fixed-model audits, 300 input rows were sampled for each corrected FA and S-SA checkpoint. Inputs were replaced group by group with either the training mean or a same-city, same-season donor value, and the change in model output defined a sensitivity ranking. The Spearman correlation between that ranking and each method's ranking, attention weights and SHAP values (128 samples), was averaged over seeds and rows. In an exploratory audit of temporal attention, 1,000 rows were used to compare replacing the days attention ranked highest against random and reverse rankings. High agreement shows that a ranking follows the model's behaviour; it does not show physical causation.

I. Pseudocode

Algorithm 1. Locked forecast evaluation
Input: aligned table D; periods T_fit, T_sel, T_cal, T_test
Output: frozen model M_final, scores P, decision
D_c ← PREPROCESS(D) // UTC day, AQI_eod, flags, lags
for each candidate m and feature family f **do**
  fit m on D_c[T_fit]; s_m ← RMSE(m, D_c[T_sel])
end for
M_final ← argmin s_m; r ← 90th percentile of |y − M_final(x)| on D_c[T_cal]
FREEZE(M_final, r) and record file hashes
ŷ ← M_final(x) on D_c[T_test]; b ← previous-day AQI on D_c[T_test]
P ← {RMSE, MAE, R^2} of ŷ and b on identical rows
for k = 1 **to** 2000 **do**
  resample calendar weeks; g_k ← RMSE(b) − RMSE(ŷ)
end for
CI ← 2.5th and 97.5th percentiles of g
promote M_final only if CI lower bound > 0 against b and the incumbent, with no sharp-change loss
return M_final, P, CI

Algorithm 2. Faithfulness audit of a frozen attention model
Input: model M, rows X, replacement rules R = {mean, seasonal donor}
Output: mean Spearman agreement of attention and SHAP
S ← SAMPLE(X, 300)
for each row x in S **do**
  a ← ATTENTION(M, x); h ← SHAP(M, x)
  for each rule r in R **do**
    for each input group q **do** δ_q ← |M(REPLACE(x, q, r)) − M(x)|
    ρ_att ← SPEARMAN(a, δ); ρ_shap ← SPEARMAN(|h|, δ)
  end for
end for
return mean ρ_att and ρ_shap over rows, averaged over seeds

IV. RESULTS AND DISCUSSION

A. Performance metrics

Each AQI-history model beat the baseline on its own rows (Table VIII) [29]. The gain was 3.39 AQI points (9.4%) for ridge with history in 2024, 2.87 (8.1%) for the GRU in 2025 and 4.40 (12.7%) for weather XGBoost in 2026, each with a 95% interval above zero. This indicates that the models forecast the next day more accurately than repeating today's value, but the typical error is still near 20 AQI points. Ridge-env, without AQI history, had RMSE 63.239, 75% above the

baseline, so inputs other than AQI history are not enough on their own. The nominal 90% ranges covered 94.7% to 95.4% of observations, so they are conservative.

TABLE VIII
LOCKED TEST RESULTS (RMSE, MAE IN AQI POINTS)

Model	Test	Rows	MAE	RMSE	R^2	Gain over baseline [95% CI]	Coverage
Baseline	2024 same-day	28,803	23.72	36.197	0.869	n/a	n/a
Ridge-env	2024 same-day	28,803	45.80	63.239	0.599	none	n/a
Ridge-history	2024 same-day	28,803	23.00	32.806	0.892	3.39 [2.03, 4.67]	94.7%
Baseline	2025 next-day	30,263	22.98	35.388	0.867	n/a	n/a
GRU-28days	2025 next-day	30,263	21.86	32.516	0.887	2.87 [2.17, 3.60]	95.4%
Baseline	2026 next-day	20,086	21.76	34.668	0.800	n/a	n/a
Frozen GRU-28days	2026 next-day	20,086	20.39	31.707	0.833	2.96	n/a
Weather XGBoost	2026 next-day	20,086	19.35	30.270	0.848	4.40 [3.19, 5.71]	94.9%

B. Comparison with baselines and ablations

A natural concern is that the GRU mostly repeats yesterday's AQI. To test this, the frozen GRU was re-run on all 2025 test rows with parts of its input removed (Table IX). Its predictions correlate 0.989 with yesterday's AQI and 68% lie within 10 points of it, so the concern is largely right: the GRU is anchored to yesterday. Yet yesterday's value alone is not enough. Given only yesterday's AQI, the GRU kept 12% of its gain; with yesterday repeated and calendar and location kept, 43%; and with the full history but no calendar or location, 55%. A ridge model on all lags, calendar and location kept 78% of the gain, so the GRU adds only about 0.6 RMSE points beyond a linear model. The gain comes from older days, season and location, and from pulling extreme values toward the average (the one-variable fit is $y = 10.5 + 0.921 \times$ yesterday). The inputs in the removal tests are unusual for the network, so they overstate its dependence on the removed data, and the linear comparisons are the cleaner evidence.

TABLE IX
INPUT ABLATIONS OF THE 2025 GRU AND LINEAR COMPARISONS

Model	RMSE	Gain	Share of GRU gain
Baseline (yesterday)	35.388	0.00	0%
Linear, yesterday only	34.856	0.53	19%
Linear, three recent inputs	33.471	1.92	67%
Linear, all history, calendar, location	33.155	2.23	78%
GRU, only yesterday's AQI	35.044	0.34	12%
GRU, yesterday repeated	34.140	1.25	43%
GRU, no calendar or location	33.794	1.59	55%
GRU, older days shuffled	33.303	2.09	73%
GRU, full model	32.516	2.87	100%

Adding inputs step by step shows the same pattern for the tabular models fitted on 2023 data (Table X): only the AQI-history step beat the baseline, and station location helped within known cities but not on an unseen one. For held-out cities, the spatial S-TA+SA model reached a city-averaged RMSE of 54.16 against 57.19 for its uniform control and improved five of eight cities, whereas sequence TA did not improve on its control (55.23 against 55.11). This is cross-city accuracy and is not evidence about faithfulness.

TABLE X
BEST TEST RMSE BY INPUT SET, 2023 TABULAR MODELS

Inputs	Chrono.	Held-out city	Season
Satellite only	119.7	72.4	80.3
Weather only	97.8	60.8	60.7
Satellite + weather	93.5	58.5	57.2
+ station location	86.8	57.1	49.7
+ day of year, past weather	72.1	57.2	49.0
+ AQI history	42.4	31.3	34.9
Baseline (yesterday)	46.7	32.8	40.1

C. Feature importance of the weather model

The weather XGBoost uses 50 inputs (Table IV) and predicts the change from yesterday. On the 2026 test rows, the forecast weather carried 28% of the model's mean absolute SHAP value, close to yesterday's AQI at 29% and ahead of the older daily values at 24% (Fig. 2). Shuffling the weather inputs raised RMSE by 7.2 points, and by 5.9 when shuffled within each city, which keeps city climate. Setting the forecast to missing raised RMSE from 30.27 to 32.23, removing about 44% of the model's gain over the baseline. Forecast wind speed and humidity mattered most (RMSE rises of 3.2 and 2.5), which agrees with the physics of dispersion and moisture, but these scores describe the model's behaviour and do not prove cause. The development rows gave the same ranking. On those rows the history-only XGBoost scored close to the baseline, while adding weather lowered RMSE by 3 to 5 points (Table XI).

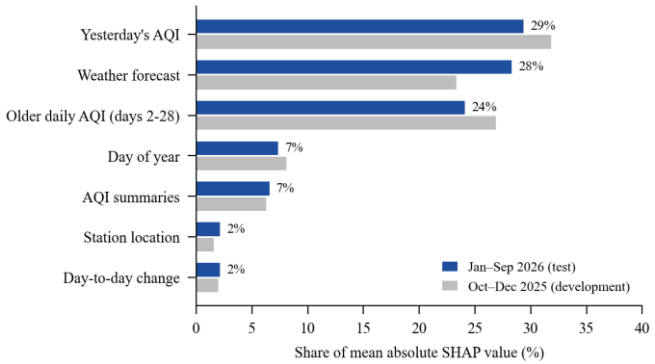


Fig. 2. Share of the weather XGBoost's mean absolute SHAP value by input group.

TABLE XI
WEATHER FORECAST ABLATION ON DEVELOPMENT ROWS (RMSE)

Model	Oct 2025	Nov–Dec 2025
Baseline	37.26	41.70
Frozen GRU	36.23	38.16
XGBoost, history only	36.78	40.25
XGBoost, with weather forecast	31.80	37.24

D. Attention and accuracy

Attention did not improve accuracy reliably (Fig. 3). Across the 36 paired comparisons, attention helped in 11, hurt in 11 and was inconclusive in 14 [30]. The largest reversal was S-SA in the chronological test, which gained 9.49 on SWT-w7 and lost 3.58 on SW-w7. The most consistent result was S-TA+SA in held-out cities, which gained 3.03 on SWT-w7 and 3.00 on SW-w7 but showed no chronological gain on either set. Feature attention alone was a clear gain once and a clear loss in four of its six comparisons. With 36 unadjusted intervals, a few clear results would be expected by chance. These models omit AQI history, so their RMSE (about 53 to 95) is far above the forecast models'.

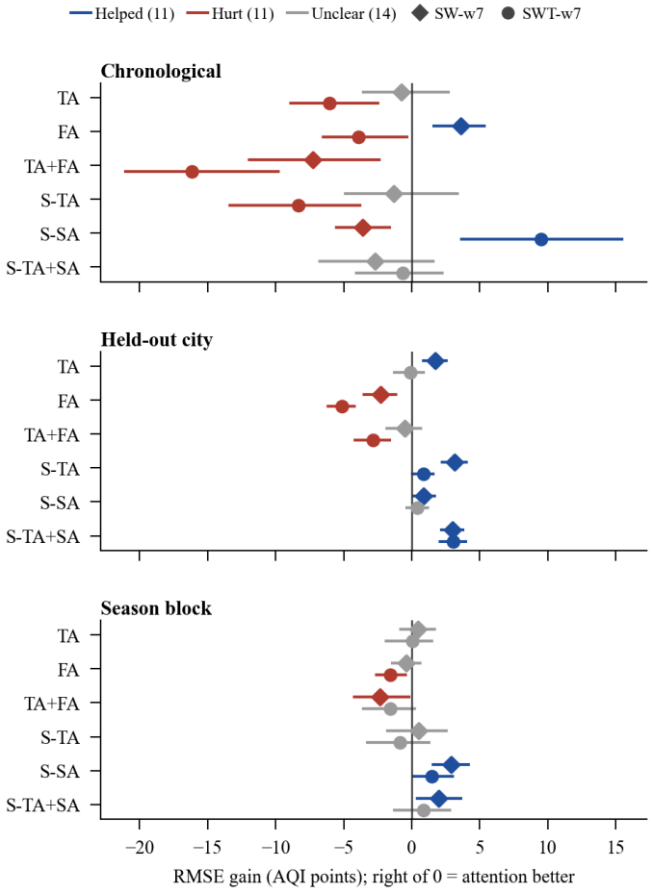


Fig. 3. RMSE gain of each attention variant over its uniform control, with 95% intervals, on two input sets.

E. Faithfulness of attention

In all four fixed-model audits attention agreed less than SHAP with replacement sensitivity (Table XII), which suggests that attention is a weaker guide to what the model uses. The result varied by city and replacement method. Under mean replacement, spatial attention nearly matched SHAP in Bengaluru (0.673 against 0.677) but was near zero or negative in Delhi (0.100), Jaipur (−0.017) and Lucknow (−0.104). City samples are uneven, from one row in Mumbai to 148 in Delhi, so small-city figures are weak.

TABLE XII
RANK AGREEMENT WITH REPLACEMENT SENSITIVITY (SPEARMAN)

Attention type	Replacement	Attention	SHAP
Feature (FA)	Training mean	0.420	0.664
Feature (FA)	Seasonal donor	0.392	0.546
Spatial (S-SA)	Training mean	0.242	0.419
Spatial (S-SA)	Seasonal donor	0.483	0.586

The exploratory temporal-attention audit shows the weights are not empty. Replacing the day attention ranked highest moved the output by 37.1 AQI points on average, against 17.3 for a random day and 7.5 for the lowest-ranked day. Rank agreement with leave-one-day-out sensitivity was 0.488 overall, and it did not fall on polluted days: 0.547 at AQI 300 or above (320 rows) and 0.461 below (680 rows). Chennai was weakest at 0.178.

F. Error analysis

All models read too high on clean days and too low on polluted days (Fig. 4). The Severe band is only 2.4% of station-days, so the models see few examples of it and regress toward the mean. In the 401–500 band (94 station-days in 2026) the mean error was −44 for the weather model and −72 for the GRU, which means none of the models is suited to alerting on the worst days. Sharp changes are also missed (Table XIII). The weather model detected 18.2% of moves of 50 points or more in the correct direction against 6.7% for the GRU, but its precision fell from 57.5% to 52.8%, so it failed the promotion rule's no-loss condition despite a lower RMSE (paired gain over the GRU 1.44, 95% interval 0.38 to 2.61). On days that changed by less than 25 points, the baseline had the lowest error, so the models earn their gain on the days that change.

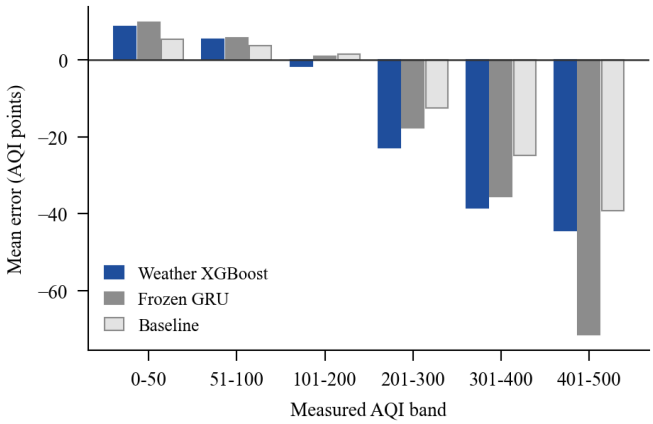


Fig. 4. Mean signed error by measured AQI band, January–September 2026. Below zero means too low.

TABLE XIII
RMSE BY SIZE OF ACTUAL DAILY AQI CHANGE, 2026

Change (points)	Days	Weather	GRU	Baseline
Under 25	14,202	15.98	13.82	11.05
25 to under 50	3,569	33.00	33.52	35.57
50 or more	2,315	68.59	76.28	87.91

By city, the GRU improved on the baseline in all eight cities in 2025 (Fig. 5), with the largest gain in Delhi (3.74) and the smallest in Mumbai (1.29), although Delhi's error stays the largest at 41.5. Two alert classifiers fit on October 2024 and scored once on 2025 did not pass their promotion checks and were not deployed: for AQI of 300 or more the GRU threshold had precision 83.7% and recall 82.9% against 88.4% and 64.9% for the classifier.

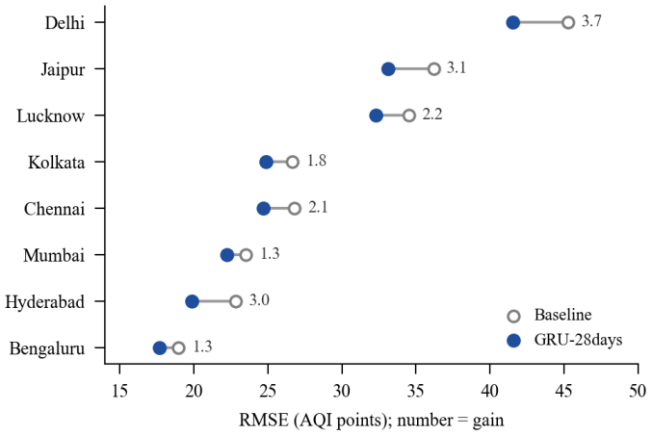


Fig. 5. RMSE of the GRU and the baseline by city, 2025 test (30,263 station-days). Numbers show the gain.

G. Interpretation of results

Three conclusions follow. First, forecast skill beyond the baseline is real but modest: 8% to 13% in RMSE. Most of what the models predict is yesterday's value, the history-only GRU adds little over a linear model, and the clearest additional signal comes from weather forecasts. Second, attention is neither a reliable accuracy gain nor, in the sampled audits, as faithful a guide to model behaviour as SHAP, and its effects depend on the input set, the test design and the city. An attention map should therefore be checked for each model and setting before it is read as an explanation. Third, the forecasts regress toward the middle, which limits their use on the days that matter most for health. These findings describe model behaviour on past data and do not identify physical causes of pollution.

H. Comparison with past literature

The reviewed studies report scores on different targets, horizons and scales, so this work does not rank its numbers against theirs (Table XIV). The comparison below is therefore about what each study shows, and it connects the results to the gaps in Section II.

TABLE XIV
REPORTED RESULTS OF REVIEWED STUDIES AND THEIR RELATION TO THIS WORK

Study	Reported result	Relation to this work
Chen et al. [6]	Hourly PM2.5, RMSE 9.82 one step and 13.49 averaged over 1–48 h	Different pollutant, horizon and scale. Attention is a predictive precedent there; here its accuracy effect varied (11 helped, 11 hurt, 14 unclear).
Liu et al. [7]	PM2.5 MAE 13.97, 16.13 and 4.07 on three datasets	Component ablation there tests architecture; here a fixed-model audit tests whether attention reflects model behaviour.
Ma et al. [5]	Lower error with four frozen layers than with one after target-site fine-tuning (scaled units)	Adaptation uses target labels. The locked tests here use none.
Li et al. [19]	Site-based PM10 R^2 about 0.85 on 288 held-out stations	Spatial assessment, not forecasting. Next-day R^2 here was 0.887 (2025) on a different target.
Liu et al. [14]	Beijing daily AQI RMSE 3.0012	Different scale and preprocessing, and split independence is unverified there.
Zhang et al. [10]	Intrinsic explanations agreed closely with GradientSHAP	Here attention agreed less than SHAP with replacement sensitivity in four of four audits. Models and cities differ, so the contrast is indicative.
Turbé et al. [8]	Shapley sampling ranked best among six attribution methods	Consistent in direction: SHAP agreed more with measured sensitivity than attention did.
Šić et al. [9]	Suitable perturbation depends on dataset and model	Consistent: S-SA agreement moved from 0.242 to 0.483 when the replacement rule changed.

V. CONCLUSION

This study set out to measure how much AQI and weather models improve next-day station forecasts over a previous-day baseline, and whether attention explains what drives such models. It built an aligned dataset for 107 stations in eight cities and ran three locked forecast tests. The GRU lowered RMSE from 35.388 to 32.516 in 2025 and the weather XGBoost from 34.668 to 30.270 in 2026, while the history-only GRU added little beyond a linear model and the weather forecast supplied much of the extra skill. Across 36 comparisons attention did not reliably improve accuracy, and in every sampled audit it agreed less than SHAP with measured sensitivity. The contribution is a reproducible evaluation that separates forecast skill, input contribution and explanation faithfulness, with a dashboard of the saved results. The findings suggest that attention maps should be tested before they are used to explain environmental forecasts, and

that a previous-day baseline should accompany any reported accuracy.

VI. Study Limitations

Dataset constraints. The study used CPCB station exports and public satellite, reanalysis and archived forecast data for eight Indian cities. Station coverage is uneven, since Delhi supplies 39 of 107 stations, and 16 stations were not scored in 2026 because exports were unavailable or had no usable targets. Satellite values are limited by cloud cover, and the archived weather forecasts are city-level with an unverified delivery delay. The attention study uses 2023 data only.

Technical and environmental restrictions. All experiments ran on a single laptop with an 8 GB GPU. The attention study used three seeds and the fixed-model audits sampled 300 rows per checkpoint (1,000 for the temporal-attention audit), so the intervals cover test-week sampling and not training-seed variation. Replacing inputs with a training mean can create unrealistic inputs, and correlated inputs share credit in importance scores.

Scope limitations. The evaluation was offline on historical data and did not include live forecasts, because no same-day AQI source is configured. The out-of-time evaluation for 13 October to 31 December 2026 has no score as of 2 November 2026. The audits are model-behaviour interventions and not causal tests of atmospheric processes. LIME, remove-and-retrain, concept-activation testing and an audit of the held-out-city checkpoints were outside the scope of this study.

VII. Future Scope

Suggested improvements. The frozen blend of 30% GRU and 70% weather XGBoost should be scored on the out-of-time evaluation after 1 January 2027 under the promotion rule stated in Section III. A history-only XGBoost can be retrained and scored on the 2026 test to isolate the weather effect cleanly. The audit can be extended to the held-out-city checkpoints and to LIME, remove-and-retrain and a concept-based stagnation test. Tail-aware training objectives could address the under-forecasting of the Severe band, and boundary-layer height could be added as a physically relevant input.

Real-time implementation. A same-day AQI source and a live weather forecast feed would allow forecasts to be recorded before their target day, so that live skill can be measured and not only backtested. The out-of-time status is tracked in [31].

Potential applications and extensions. The method can be extended to PM2.5 and PM10 targets, to more cities, and to health advisories built on the dashboard. The audit procedure can also be reused to test attention explanations in other environmental forecasting tasks.

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